Aligning LLMs with Biomedical Knowledge using Balanced Fine-Tuning

Zhenchao Tang*[1], Fang Wang*†[1], Haohuai He[1], Jiale Zhou[1], Tianxu Lv[1], Jun Zhu[1], Shouzhi Chen[1], Minghao Yang[1], Yu Wang[1], Jiayang Wu[1], Yidong Song[2], Yaokun Li[3], Jiehui Huang[4], Jun Zhou[5], Bing He[1], Jianhua Yao†[1]

[1]Tencent AI for Life Sciences Lab [2]Nankai University [3]Sun Yat-sen University
[4]The Hong Kong University of Science and Technology [5]Tsinghua University
*Equal contribution †Correspondence: avonwanghit@gmail.com, jianhua.yao@gmail.com

NEURAL INFORMATION PROCESSING SYSTEMS

Tencent AI for Life Sciences Lab

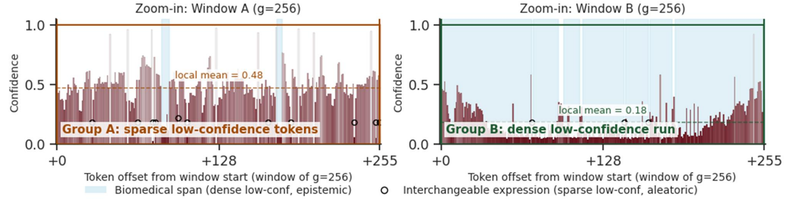
NeurIPS 2026 arxiv.org/abs/2511.21075
Project and code: shapsider.github.io/BFT

1 Problem & challenges

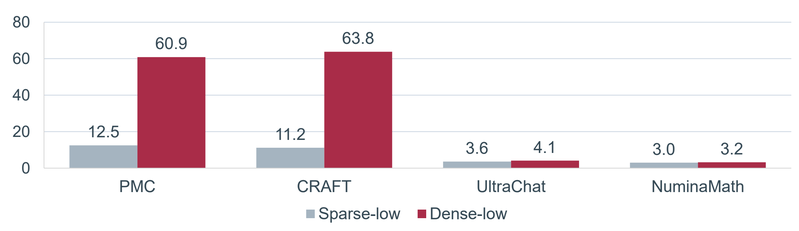
Biomedical alignment must retain rare entities and mechanistic reasoning under limited supervision. DFT weights token losses by absolute confidence, which can suppress entire difficult regions. Sparse outcome rewards further make a useful supervised initialization essential.

Observation: uncertainty has local structure

Teacher-forced PMC responses, base 70B model, g = 256



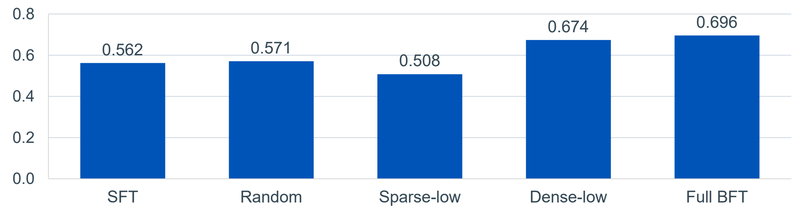
Share of biomedical entities / expert concept spans (%)



Dense biomedical runs are enriched for concepts. CRAFT uses expert annotations, while general-domain controls show little entity enrichment. This supports a knowledge-gap interpretation without directly labeling epistemic uncertainty.

Retention control: selected regions matter

VCWorld accuracy, 70B, same supervised training recipe



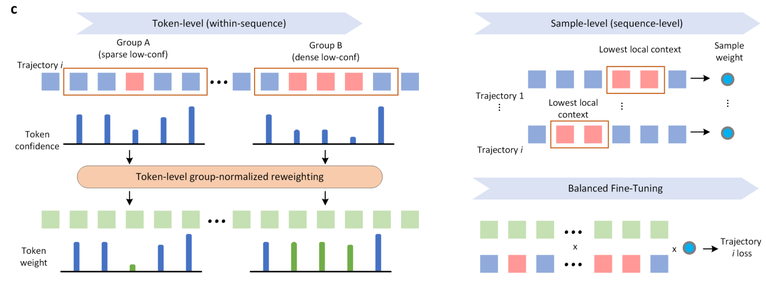
Random retention matches the dense-window token budget. Dense-only retention reaches 0.674 vs 0.571 for random selection. Full BFT replaces discrete masks with continuous weights.

2 Motivation

A token's local context should determine how much learning signal it retains. Difficult biomedical runs need stronger token signals, and responses containing those runs need more training budget. BFT implements these complementary decisions with one shared local-confidence window.

3 Method: Balanced Fine-Tuning

BFT computes each target probability c and its centered-window mean C^{loc}. Relative difficulty sets token weights. The weakest local region sets the sample weight.



Token level: normalize by local confidence

$\rho_{b,t} = \mathrm{clip}(c_{b,t} / (C_{b,t}^{loc} + \varepsilon), 0, 1)$

Context	c	C^{loc}	DFT weight	BFT ρ
Sparse-low	0.10	0.87	0.10	0.115
Dense-low	**0.10**	**0.20**	**0.10**	**0.50**

Sample level: emphasize the weakest region

$s_b = \mathrm{sg}(1 - \min_t C_{b,t}^{loc})$

The bounded s_b coefficient reallocates budget across responses, while token weights redistribute it within each response. Both signals use stop-gradient.

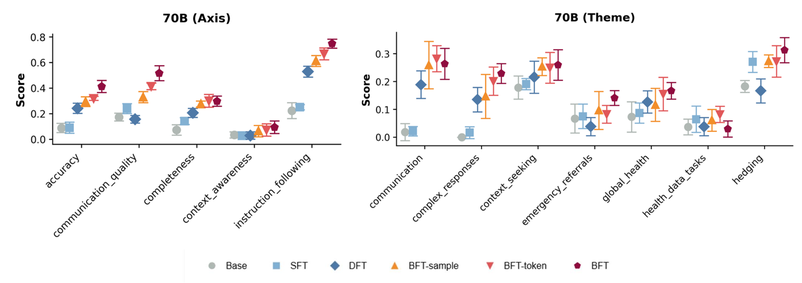
Joint response-token objective

$$L = (1/B) \Sigma_b s_b [\Sigma_t m_{b,t} \mathrm{sg}(\rho_{b,t}) \ell_{b,t}] / [\Sigma_t m_{b,t} + \varepsilon]$$

$\ell = -\log c$. m selects valid response targets. C^{loc} averages a boundary-aware centered window. g = 256 is the only added tunable hyperparameter, with $\varepsilon = 10^{-8}$.

Shared recipe: 48,503 examples, two full-parameter epochs, learning rate 10^{-5}, 4096-token cutoff. We evaluate reasoning and transfer under this matched setup.

HealthBench Hard (70B) & GeneAgent



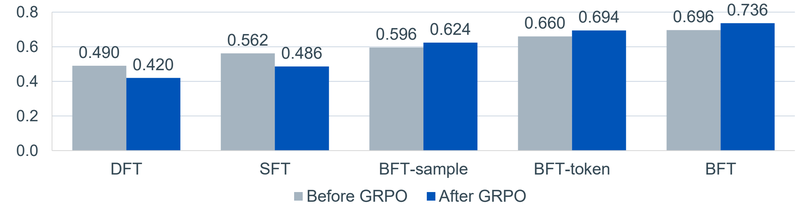
BFT improves more consistently across 14B / 32B / 70B and has the best overall GeneAgent ROUGE among aligned open-weight models. Error bars denote decoding variability.

4 Experimental results

Matched SFT / DFT / BFT backbones and token-only / sample-only ablations. Agent pipelines stay fixed while the language-model backbone changes.

Chemical perturbation reasoning + GRPO

70B Up/Down/No accuracy, five held-out cell lines
Sparse-reward GRPO uses 45 training cell lines

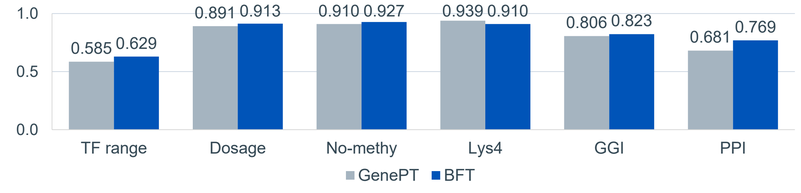


+13.4 pp over SFT before GRPO
+4.0 pp after GRPO, while SFT / DFT decline

Token-only and sample-only gains are intermediate. BFT vs Gemini-2.5-Flash is not significant: +0.018, 95% CI [−0.004, 0.040], p = 0.088.

Profile embeddings: gene-level AUC

BFT profiles use text-embedding-ada-002. Expression-weighted gene embeddings also produce cell representations.

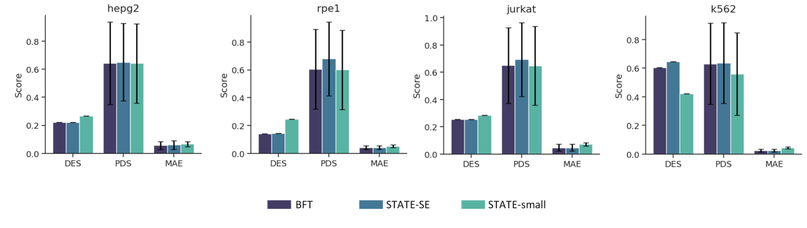


Mean gene AUC: 0.829 vs 0.802. GenePT remains higher on Lys4. The encoder is matched, while profile source and construction differ.

Cell-type clustering: ARI / AMI / ASW

Method	ARI	AMI	ASW
scGPT	0.26	0.32	0.039
GenePT	0.30	0.46	0.041
SFT	0.12	0.25	0.021
BFT	**0.34**	**0.58**	**0.061**

Perturbation-response prediction



Comparable to standard STATE inputs across four datasets. Profile generation is zero-shot for these datasets, while the STATE decoder remains trained.